Size Confounds Semantics: An Audit of SING's Null-Space Leakage Score

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Workshop submission

Abstract

SING [10] ranks classifiers by how much class-relevant semantics leak into the linear head's null space. Its standout result is that a "DinoViT" pretrained with DINO leaks least. We audit that claim. Every public ViT-L/16 checkpoint consistent with SING's code or wrappers turns out to be supervised or SWAG-supervised; none matches the reported 84.0% top-1, and none is DINO. Within a fixed model, SING's absolute-shift score |AS| grows by a factor of ≈ 10 as the removed null subspace grows from d=24 to d=1048 on 3 of 4 models. Even at matched d=24, the ranking disagrees with SING's published order (Kendall τ=0.29). A pooled log-linear fit gives $|\mathrm{AS}| \propto \rho^{0.43}$ in removed feature energy ρ, close to a simple $\sqrt{\rho}$ geometric prediction, but the separation of energy from dimension is mixed rather than decisive. We define a size-fair score, $\mathrm{AS}_{\mathrm{fair}}$, which divides null removal by an energy-matched random-subspace baseline. On all 13 SING models, $\mathrm{AS}_{\mathrm{SING}}$ reproduces the published Fig. 6b ordering (τ=0.82) and $\mathrm{AS}_{\mathrm{fair}}$ breaks it (τ=−0.18); two class-level claims reverse with bootstrap CIs. Planted-leakage recovery is weak for both scores. When null-space scores are used to compare architectures, we recommend reporting ρ and $\mathrm{AS}_{\mathrm{fair}}$ alongside raw AS.

1 Introduction

A linear classifier on features $f \in \mathbb{R}^m$ has a null space of dimension $m-c$ (for c classes) that leaves the logits unchanged. SING [10] treats class-relevant content in that null space as "leakage": information the head could have used, but did not. Its Absolute Shift (AS) measures how far a CLIP-aligned [5] translation of f moves once the null component is removed. SING uses AS (and the IS/AS ratio) to rank 13 ImageNet models and singles out a ViT labeled "DinoViT" as the least leaky [1].

Null-space analyses also appear in OOD detection [2, 9] and in overfitting scores [6]. Across architectures, though, the null dimension $d=m-\mathrm{rank}(W)$ and the null energy fraction $\rho = \mathbb{E}\|V_n V_n^\top f\|^2 / \mathbb{E}\|f\|^2$ both vary by orders of magnitude. If AS mostly tracks how much energy is removed, then cross-model comparisons of "semantics" are confounded by size.

Contributions. (1) We find no way to identify SING's "DinoViT" as DINO: every matching candidate is supervised, and none reproduces 84.0% top-1. (2) Within-model AS(d) curves rise by $\approx 10\times$ from d=24 to d=1048 on ResNet50, ResNeXt101 and DINOv2-L. BiT [3] is the exception, and translator cancellation explains it. (3) Energy and dimension are hard to separate. A pooled fit gives exponent a=0.43 on ρ, against 0.5 from a $\sqrt{\rho}$ prediction, but flatness at fixed ρ is not clean. (4) On 13 models, the size-fair $\mathrm{AS}_{\mathrm{fair}}$ breaks SING's published ranking (τ: 0.82 → −0.18) and reverses several claims, subject to the caveats we state. We offer $\mathrm{AS}_{\mathrm{fair}}$ as a reporting correction, not as a new ground-truth leakage oracle.

2 Background and the DinoViT identity

SING AS in brief. Let $T(f) = Af + b$ be SING's released linear translator into CLIP space and V_n an orthonormal basis of $\ker(W)$ (tolerance 10^{-5}). For a text direction t, SING's per-image absolute shift is the change in angle $\theta(T(f), t)$ after removing $V_n V_n^\top f$. We write $\mathrm{AS}_{\mathrm{SING}}$ for the image-averaged |AS| under full null removal (class text unless noted).

Exp 0b: "DinoViT" is not DINO. SING reports DinoViT at 84.0% ImageNet top-1 (Supp. Table 2). The released models.yaml loads a timm AugReg ViT-L/16, while the wrapper in the supplement uses torchvision ViT attribute names. We ran five public ViT-L/16 candidates on the full 50k validation set (native and force-224): torchvision IMAGENET1K_V1 (79.67%), timm AugReg (85.85%), SWAG linear [7] (85.14%), SWAG E2E (88.07%), and HF google/vit-large-patch16-224 (81.98%). None falls in 84.0 ± 0.3 (Fig. 1). All of them are supervised or SWAG weakly supervised, and DINO never released a ViT-L with a 1000-way head [1]. So the headline attribution to self-supervised DINO does not hold for any checkpoint we can identify.

Exp 0 (re-analysis). We digitized the IS/AS bars in SING Supp. Fig. 6b and correlated them with measured null dimension over 13 models. This gives Spearman −0.46 (p=0.12), with DenseNet null dim = 27 at tol 10^{-5}. Null energy on its own predicts much better:

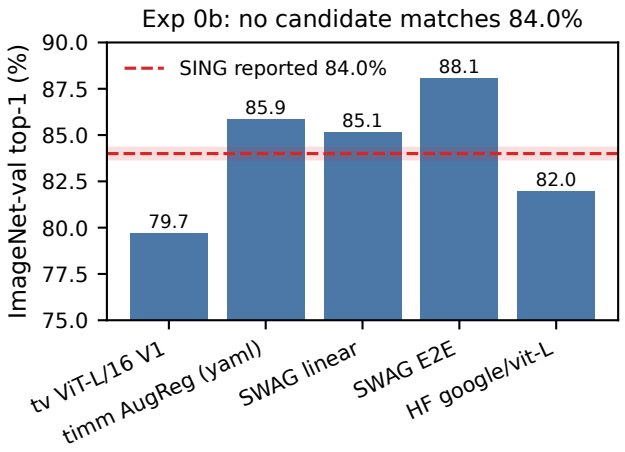


Figure 1: Exp 0b top-1 on ImageNet-val. No candidate matches SING's reported 84.0%.

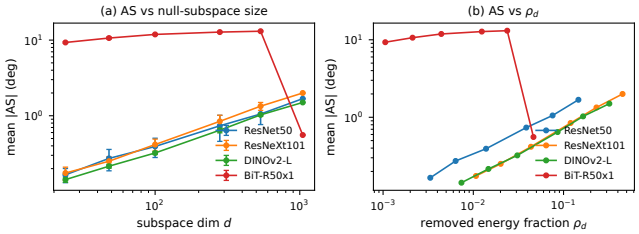


Figure 2: Exp 1: mean |AS| vs. subspace size d (a) and removed energy ρ_d (b). BiT collapses at the full null (translator cancellation).

Spearman(ρ_{full}, IS/AS) $= -0.94$ (p=1.9×10^{-6}). DeiT3-L [8] has null dim 24 yet the worst ratio. Dimension alone cannot explain the ranking, and that is why we turn to a within-model test.

3 Within-model size dependence (Exp 1)

We fix a 10k ImageNet-val subset (10 images/class) and remove random d-dimensional subspaces of each model's own null space ($d \in \{24, 48, 100, 280, 536, 1048\}$, 20 seeds for d<d_{null}). The models are ResNet50, ResNeXt101, BiT-R50x1 and DINOv2-L (all d_{null}=1048), with SING's released translators.

Fig. 2 and Table 1 show r=AS(1048)/AS(24) ≈ 10 on 3 of 4 models. With d=24 matched across 8 models, Kendall τ against the Fig. 6b order is 0.29 (p=0.40) for |AS|. Both pre-registered GO conditions hold.

BiT anomaly. BiT's full-null AS collapses (r=0.06), even though partial subspaces give large AS (∼9–13°). The singular values of AV_n are extreme (top 284 vs. 5.6 for ResNet50). When the orthogonal null blocks are summed, they nearly cancel through the translator, so full-null AS understates how sensitive BiT is per direction. This comes from translator conditioning; it is not a code bug.

Model	r	2σ band	R^2 vs d	R^2 vs ρ_d
ResNet50	10.15	[5.83, 14.48]	0.983	0.983
ResNeXt101	11.44	[6.93, 15.95]	0.977	0.977
DINOv2-L	10.48	[8.85, 12.10]	0.974	0.974
BiT-R50x1	0.060	[0.057, 0.063]	0.512	0.512

Table 1: Exp 1: r=AS(1048)/AS(24). Energy and dimension fits are collinear for random subspaces.

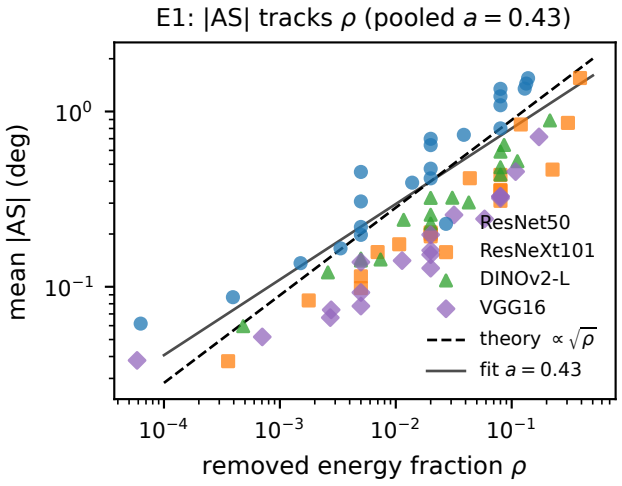


Figure 3: E1 cells: |AS| vs. ρ. Dashed: $\sqrt{\rho}$ theory; solid: fitted a=0.43.

Caveat. For random subspaces ρ_d is nearly linear in d, which makes R^2 vs. d and vs. ρ_d identical (≈ 0.98). Exp 1 by itself cannot tell energy from dimension.

4 Energy vs. dimension (E1): a mixed result

Geometry. Suppose a linear translator maps a removed feature component of relative energy ρ into CLIP space. When the residual is small relative to $\|T(f)\|$, the first-order angular displacement scales with the removed magnitude $\|\Pi f\| \propto \sqrt{\rho}$. That predicts |AS| $\propto \rho^{1/2}$ at fixed geometry, i.e. an exponent of 0.5.

Design. On ResNet50, ResNeXt101, DINOv2-L, VGG16 and EVA02-L we (a) compare high-energy with low-energy null PCs at fixed $d \in \{24, 100, 280\}$, and (b) hit fixed $\rho \in \{0.5\%, 2\%, 8\%\}$ at several d by mixing top and bottom PCs. The pooled fit over 76 feasible cells,

$$\log|\text{AS}| \;=\; a\log\rho \;+\; b\log d \;+\; c_{\text{model}},$$

gives a=0.431 [0.430, 0.433], b=0.010 [0.008, 0.012], R^2=0.93 (Fig. 3).

At fixed d, the high/low energy ratio is above 3× on all 4 evaluable models (e.g. ResNet 11–22×, ResNeXt

$\approx 10\times$, DINOv2 3.7–5.1×, VGG16 6–10×). At fixed ρ, however, the max/min across d is above 1.5 in several cells (ResNet up to 2.29 at ρ=0.5%). For EVA02-L (d_{null}=24), nothing beyond the full-null cell can be evaluated. We therefore call E1 mixed, not decisive. Energy dominates the fit and its exponent is within ≈ 0.07 of $\sqrt{\rho}$, but the pre-registered ENERGY-DRIVEN flatness rule is not cleanly met.

5 A size-fair score

For the energy-matched random baseline, we draw a Haar subspace of the full feature space that removes exactly the same energy $\mathbb{E}\|\Pi f\|^2$ as the null projector, and take AS_{rand} as its mean |AS| (20 draws). Then

$$\text{AS}_{\text{fair}} = \frac{\text{AS}_{\text{null}}}{\text{AS}_{\text{rand}}}.$$

A value near 1 means the null shifts AS no more than a random subspace of equal energy would. Values $\gg 1$ mean the leakage is concentrated in the null. This is the definition registered for E3, and we reuse it at α=0 for the E2 audit.

6 Planted leakage (E3)

We plant a CLIP concept direction (red/striped/water) into the null spaces of 8 models at log-uniform $\alpha \in [0.01, 1]$ over 50 trials. The logits stay unchanged (max $|\Delta\text{logit}| \leq 2.3\times10^{-5}$). Kendall τ against the planted-α ranking, averaged over concepts, is AS_{SING} 0.386 $[0.324, 0.447]$ and AS_{fair} 0.071 $[0.004, 0.138]$. Neither score recovers planted strength well, and AS_{fair} is close to chance. The per-concept means are similar (red 0.386/0.070, striped 0.403/0.061, water 0.369/0.083). Some raw observations. DeiT3-L signed AS is negative at every α, and its |AS| shrinks as α grows. BiT AS_{SING} reaches 17–20° at α=1, while its AS_{fair} at α=0 is only 0.045–0.063. For EVA02-L and ConvNeXt-B, AS_{fair} is non-monotonic in α. We report this result as it came out: size-fairness and recovering a planted ranking are different goals. We propose AS_{fair} only as a size correction for comparing AS across models. It is not a calibrated leakage oracle.

7 Audit of SING's 13-model claims (E2)

On the same 10k subset, plus a 200-image claim-class set (all 50 val images of jellyfish, porcupine, Arabian camel and sports car), we recompute AS_{SING} and AS_{fair} for all 13 released translators.

Ranking. Against the digitized Fig. 6b IS/AS order, $\tau(\text{AS}_{\text{SING}})$=0.821 ($p$=1.3×10^{-5}) and

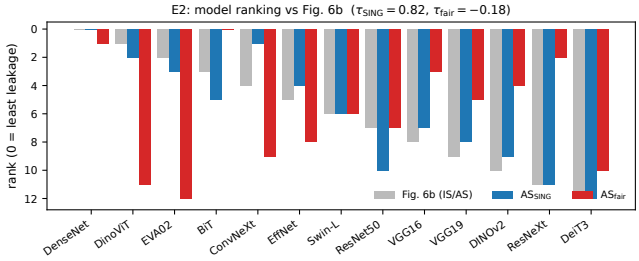


Figure 4: Ranks (0=least leakage). AS_{SING} tracks Fig. 6b; AS_{fair} does not.

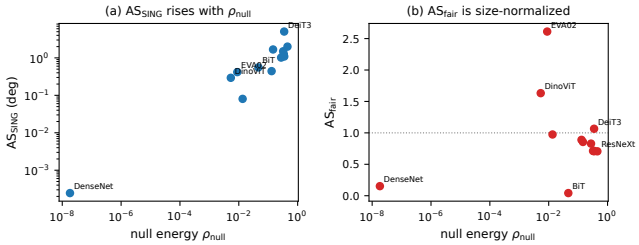


Figure 5: (a) AS_{SING} rises with ρ_{null}. (b) AS_{fair} removes that trend.

$\tau(\text{AS}_{\text{fair}})$=−0.179 ($p$=0.435). Leaving out DenseNet gives $\tau(\text{AS}_{\text{fair}})$=−0.364. Averaged over the 173 SING concepts, τ=0.641 vs. −0.026. With AS_{SING} our setup reproduces SING. The disagreement only appears after the size correction (Figs. 4–5).

Claim reversals. Under AS_{fair}, claims C1–C7, C9, C11 and C12 reverse relative to AS_{SING}: DinoViT is no longer the least leaky, and ResNeXt is no longer among the worst. Two claims reverse under both scores, with bootstrap CIs:

- C8 (ResNet porcupine / sports-car ≥90th percentile): the porcupine percentile is 84.8 $[80.1, 87.7]$ under AS_{SING} and 68.4 $[64.5, 71.5]$ under AS_{fair}; for sports car it is 61.0 $[56.1, 66.2]$ and 67.2 $[63.6, 71.4]$.
- C10 (DinoViT jellyfish ≫ camel): the jellyfish−camel CI is $[-0.205, -0.010]$ under AS_{SING} and $[-1.403, -0.458]$ under AS_{fair}, so the sign is the opposite of SING's claim.

Caveats. (i) C1/C2 are IS/AS-ratio claims, which we test here on |AS|. C11 is a steering claim, proxied by null-removal AS. C10 uses class-text AS on 50 images/class, whereas SING used ≈30 concepts / ≈1300 images. (ii) DenseNet has ρ_{null}=1.8×10^{-8}, so its AS_{fair} is degenerate. (iii) C13 (EVA02 as target space) cannot be tested, because no EVA02-space translators were released. (iv) Our DinoViT IS/|AS| is 20.95, against 70.16 in Fig. 6b. The absolute ratios differ, but the AS_{SING} ranking still agrees.

Under AS_{fair}, DinoViT lands near the worst end (12th of 13 by class AS_{fair}; EVA02 is worst). Under AS_{SING} it

ID	Claim (short)	AS_{SING}	AS_{fair}
C1	DinoViT best IS/AS among 5	holds	reverses
C2	ResNeXt lowest IS/AS among 5	holds	reverses
C3	ResNet leaks more than DinoViT	holds	reverses
C4	DinoViT least class leakage	holds	reverses
C5	ResNeXt high AS (top-2 of 5)	holds	reverses
C6	ResNet per-class AS > DinoViT	holds	reverses
C7	DinoViT per-class |AS|<1	holds	reverses
C8	Porcupine/sports-car ≥90th pct.	reverses	reverses
C9	No ResNet–DinoViT class corr.	holds	reverses
C10	Jellyfish ≫ camel (DinoViT)	reverses	reverses
C11	DinoViT least steerable (proxy)	holds	reverses
C12	Fig. 6b 13-model order	holds	reverses
C13	EVA02 target-space order	untestable	untestable

Table 2: E2 claim audit. C1/C2/C11 use |AS| proxies; C10 is class-text on 50 imgs/class.

ranks 3rd. BiT becomes the best model under AS_{fair} (0.042), and the reason is that random subspaces of matched energy move its AS by 13.2°. This is the same translator-cancellation regime we saw in Exp 1. DenseNet's near-zero ρ makes AS_{fair} numerically unstable, so we also report τ without it (-0.364).

8 Discussion

Within a single model, SING's AS is a useful probe of null content as seen through the translator. As a cross-model score, however, it mixes up semantics with how large and how energetic the null is. The boring baseline already pointed this way: over 13 models, Spearman(ρ_{full}, $|AS|_{full}$)=0.87 and Spearman(ρ_{full}, IS/AS)=-0.94. On top of that, no public checkpoint supports the DinoViT label.

What AS_{fair} is and is not. It is a reporting correction. Dividing null AS by an energy-matched random baseline stops models from being rewarded for having a small or low-energy null. It is not a validated ground-truth measure of leakage. E3 shows that neither score recovers planted α rankings at the pre-registered $\tau \geq 0.8$ bar, and AS_{fair} does worse at that particular task. We do not claim any replacement score beyond this size correction.

Recommendations. For any comparison of null-space leakage:

1. Name the exact checkpoint and its measured top-1, and avoid labels that cannot be verified, such as "DinoViT".
2. Report d_{null} and ρ_{null} next to the score.
3. Report AS_{fair} (or another energy-matched baseline) beside raw AS.
4. When full-null and block-wise AS disagree (as for BiT), flag the translator's condition number.
5. When d and ρ vary widely, prefer within-model interventions (subspace size, planted signal) to league tables across architectures.

Limitations. E1 is mixed rather than decisive on energy vs. dimension. E3 shows that AS_{fair} does not recover planted α. Several E2 claims rely on proxies (C1/C2/C10/C11), DenseNet's AS_{fair} is degenerate, and C13 cannot be tested. Because the features are derived from ImageNet, their redistribution is restricted. We did not retrain translators or evaluate a true DINO ViT-B/16 [1]. Nor did we extend the size audit to the ViM [9], Cook [2] or Rezaei [6] scores [4].

9 Conclusion

With the released translators, SING's AS ranking of 13 classifiers is reproducible, but it is confounded by size. Correcting for null energy with AS_{fair} breaks the published order (τ: 0.82 → -0.18) and reverses multiple claims, and the "DinoViT" result cannot be pinned on DINO either. Until there is a calibrated leakage oracle, cross-model null-space comparisons should report ρ and a size-fair baseline next to raw AS.

Reproducibility. Analysis code and aggregated tables are at https://github.com/sudeshgm/ic014-sing-null-dimension. The ImageNet features are private.

References

[1] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In ICCV, 2021.

[2] Matthew Cook, Alina Zare, and Paul Gader. Outlier detection through null space analysis of neural networks. arXiv preprint arXiv:2007.01263, 2020.

[3] Alexander Kolesnikov, Lucas Beyer, Xiaohua Zhai, Joan Puigcerver, Jessica Yung, Sylvain Gelly, and Neil Houlsby. Big transfer (BiT): General visual representation learning. ECCV, 2020.

[4] Mikael Makonnen, Moritz Vandenhirtz, Sonia Laguna, and Julia E. Vogt. Measuring leakage in concept-based methods: An information theoretic approach. arXiv preprint arXiv:2504.09459, 2025.

[5] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In ICML, 2021.

[6] Saeed Rezaei and Mohammad Sabokrou. Quantifying overfitting: Evaluating neural network performance through analysis of null space. arXiv preprint arXiv:2305.19424, 2023.

[7] Mannat Singh, Laura Gustafson, Aaron Adcock, Vinicius de Freitas Reis, Bugra Gedik, Raj Prateek Kosaraju, Dhruv Mahajan, Ross Girshick, Piotr Dollár, and Laurens van der Maaten. Revisiting weakly supervised

pre-training of visual perception models. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 804–814, 2022. arXiv:2201.08371; SWAG.

[8] Hugo Touvron, Matthieu Cord, and Hervé Jégou. DeiT III: Revenge of the ViT. In ECCV, 2022.

[9] Haoqi Wang, Zhizhong Li, Litong Feng, and Wayne Zhang. Vim: Out-of-distribution with virtual-logit matching. In CVPR, 2022.

[10] Harel Yadid, Guy Levi, Ron Betser, and Guy Gilboa. Make it SING: Analyzing semantic invariants in classifiers. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. arXiv:2603.14610.

A Additional numerical detail

Exp 1 full-null anchors. Mean |AS| at full null: ResNet50 1.682°, ResNeXt101 1.999°, BiT 0.556°, DINOv2-L 1.499°; at d=24: 0.166, 0.175, 9.30, 0.143 (seed SD in RESULTS.md).

E2 per-model class scores. Table 3 lists ρ_{null}, AS_{SING}, AS_{rand} and AS_{fair} used for the ranking.

Model	ρ_{null}	AS_{SING}	AS_{rand}	AS_{fair}
VGG16	0.350	1.088	1.537	0.708
VGG19	0.339	1.256	1.755	0.716
DenseNet121	1.8e-8	0.00025	0.0016	0.152
ResNet50	0.146	1.676	1.960	0.855
DinoViT	0.0054	0.294	0.180	1.632
EffNet-B0	0.130	0.443	0.498	0.889
BiT-R50x1	0.046	0.557	13.24	0.042
ResNeXt101	0.451	1.999	2.829	0.707
ConvNeXt-B	0.0135	0.080	0.083	0.974
Swin-L	0.274	1.019	1.229	0.829
DINOv2-L	0.322	1.499	2.105	0.712
DeiT3-L	0.348	5.048	4.738	1.065
EVA02-L	0.0089	0.417	0.160	2.612

Table 3: E2 class-text scores on the 10k subset (from e2_models.csv).

E3 per-concept τ. AS_{SING} / AS_{fair}: red 0.386 / 0.070; striped 0.403 / 0.061; water 0.369 / 0.083.